

Cognitive Science — Module 06: Learning — Study Questions

1. What is the difference between perception and learning in Active Inference?
2. What does it mean to say learning operates on a "slower timescale" than perception?
3. Explain the A matrix in a generative model. What does it represent, and how does learning update it?
4. Explain the B matrix. What does it represent, and how does learning update it?
5. Give an everyday example of learning that updates the A matrix (new observation-state association).
6. Give an everyday example of learning that updates the B matrix (new state-transition pattern).
7. What is Bayesian Model Reduction (BMR)? Explain in simple terms.
8. Why is simplifying a model sometimes better than making it more complex?
9. How does the accuracy-complexity trade-off (from Module 04) apply to learning?
10. Describe the three stages of skill learning (beginner, intermediate, expert) in terms of prediction error.
11. What changes in the generative model as a skill becomes automatic ("muscle memory")?
12. What role does slow-wave sleep play in learning?
13. What role does REM sleep play in learning?
14. Why does pulling an all-nighter before an exam usually backfire?
15. How does a child's learning of animal categories illustrate structure learning?
16. What is the difference between learning new information and forgetting old information? Can forgetting be adaptive?
17. How does curiosity drive learning? Connect this to the epistemic value from Module 05.
18. Can agents "unlearn" something? Give an example and explain how it works.
19. How is learning different from memorization? Which does Active Inference emphasize?
20. Compare Active Inference's view of learning with the behaviorist view (stimulus-response conditioning). What are the key differences?